Chi Square HW

Problem 1: “Is gender associated with political party?”

-Hypothesize

Test of independence

H0: There is no association between gender and political affiliation

HA: There is association between gender and political affiliation

a: 0.05

-Prepare

- Random & independent are assumed
- Expected value > 5 chisq.test(Gender_PoliticalAff)$expected
- Test statistic used chi square

-Compute

chisq.test(Gender_PoliticalAff)

p-value = 0.6492 > a(0.05) “Fail to reject null”

-Interpret

A chi-square test of independence was conducted to examine the relationship between gender and political affiliation. The results of the chi-square test were not statistically significant, p = 0.6492, which is greater than the significance level of α = 0.05. Therefore, we

fail to reject the null hypothesis. There is insufficient evidence to conclude that gender and political affiliation are associated.

Problem 2: “Is there an association between the level of education and their beliefs about the theory of evolution?”

-Hypothesize

Test of independence

H0: There is no association between beliefs and education level

HA: There is association between beliefs and education level

a:0.05

-Prepare

- Random & independent are assumed
- Expected value > 5 chisq.test(Belief_EduLevel)$expected
- Test statistic used chi square

-Compute

chisq.test(Belief_EduLevel)

p-value = less than 2.2e-16 < a(0.05) “Reject null”

-Interpret

A chi-square test of independence was conducted to examine the relationship between level of education and beliefs about the theory of evolution. The results of the chi-square test were statistically

significant, $p < 2.2 \times e^{-16}$, which is less than the significance level of $\alpha = 0.05$. Therefore, we reject the null hypothesis. There is sufficient evidence to conclude that level of education is significantly associated with beliefs about the theory of evolution.

Problem 3: "Are all colors equally likely for Milk Chocolate M&M's?"

-Hypothesize

Test for goodness of fit

H0: All milk chocolate colors are equally likely

HA: At least one differs from the majority

a:0.05

-Prepare

- Random & independent are assumed
- Expected value > 5 chisq.test(MM)$expected
- Test statistic used chi square

-Compute

chisq.test(MM)

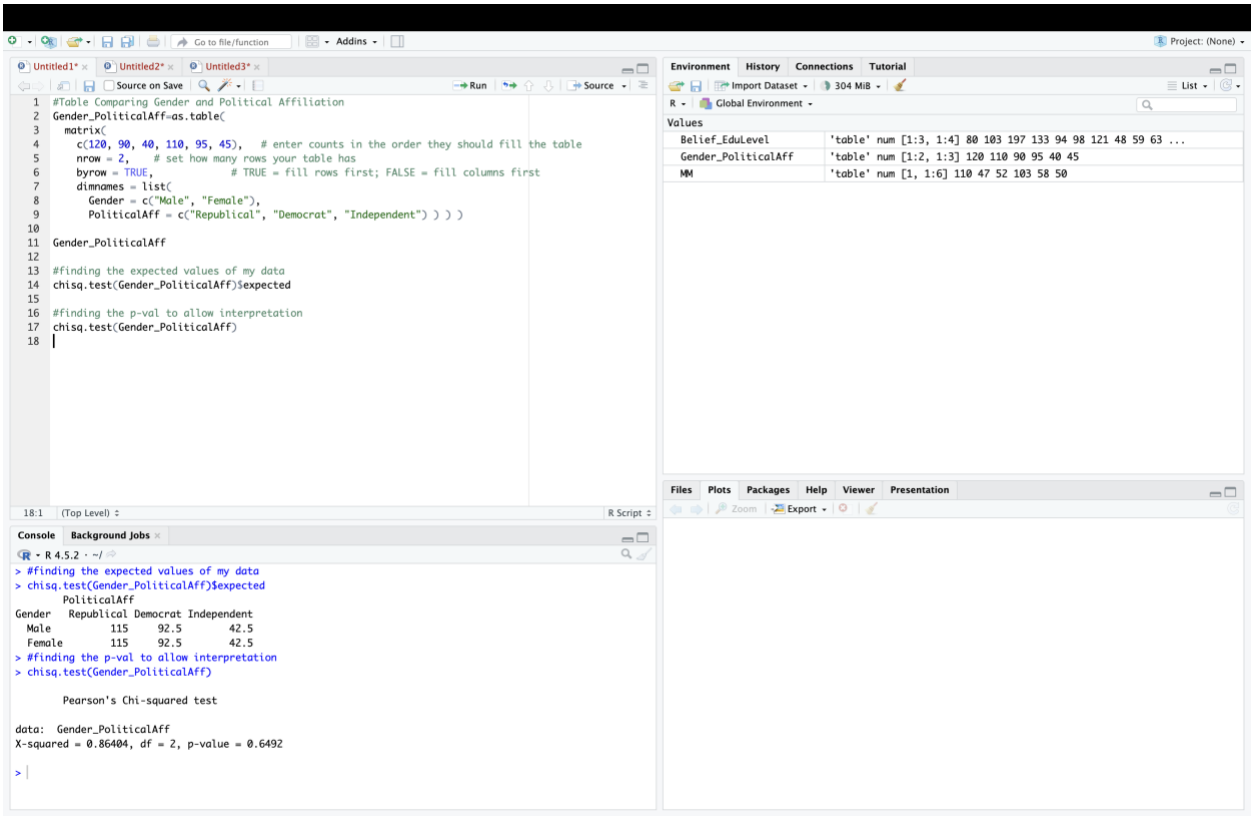
p-value = 2.637e-11 < a(0.05) "Reject null"

-Interpret

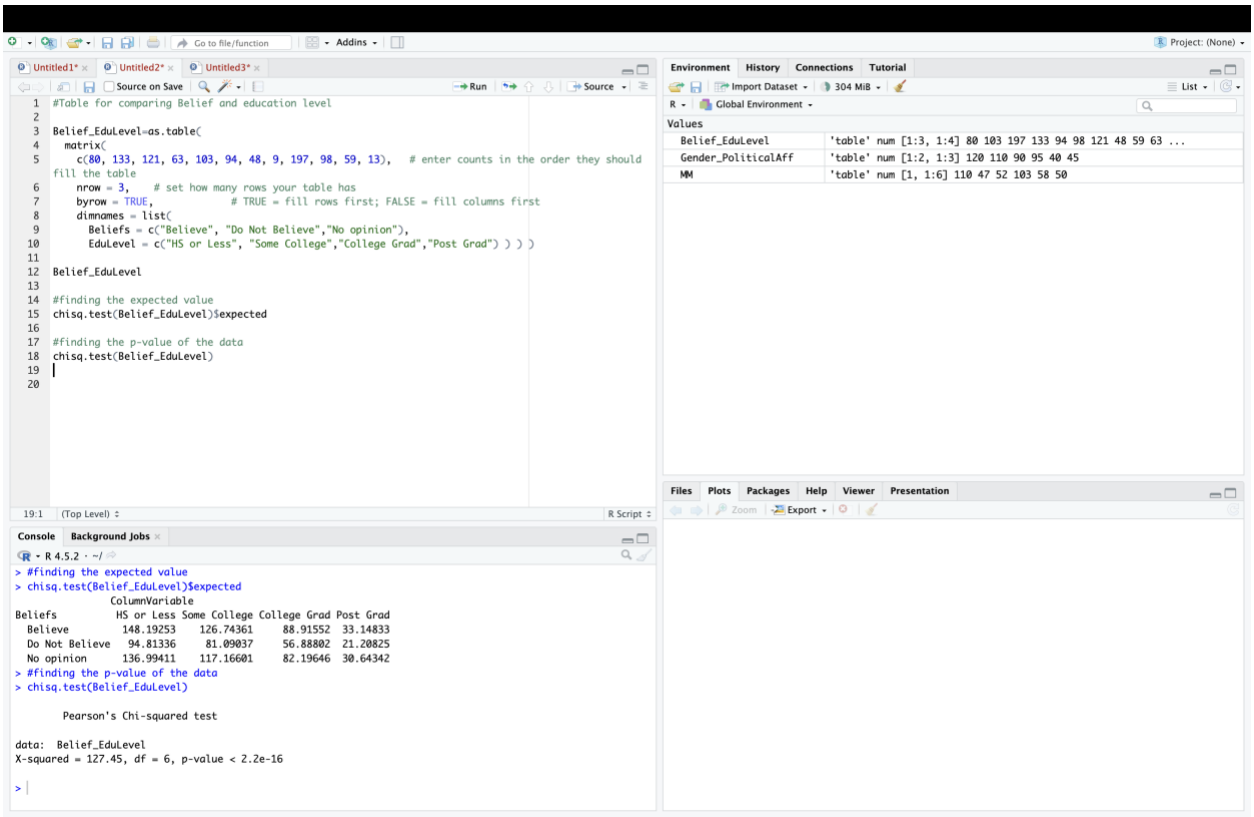
A chi-square goodness-of-fit test was conducted to determine whether the colors of milk chocolate M&Ms are equally distributed. The results of the chi-square test were statistically significant, $p = 2.637 \times e^{-11}$, which is less than the significance level of $\alpha = 0.05$. Therefore, we reject the null hypothesis. There is sufficient evidence to conclude that the colors of milk chocolate M&Ms are not equally likely.

Rstudio Code

Problem 1:



Problem 2:



```
#Table for comparing Belief and education level

Belief_EduLevel=as.table(
  matrix(
    c(80, 133, 121, 63, 103, 94, 48, 9, 197, 98, 59, 13),   # enter counts in the order they should fill the table
    nrow = 3,    # set how many rows your table has
    byrow = TRUE,             # TRUE = fill rows first; FALSE = fill columns first
    dimnames = list(
      Beliefs = c("Believe", "Do Not Believe","No opinion"),
      EduLevel = c("HS or Less", "Some College","College Grad","Post Grad") ) ) )

Belief_EduLevel

#finding the expected value
chisq.test(Belief_EduLevel)$expected

#finding the p-value of the data
chisq.test(Belief_EduLevel)
```

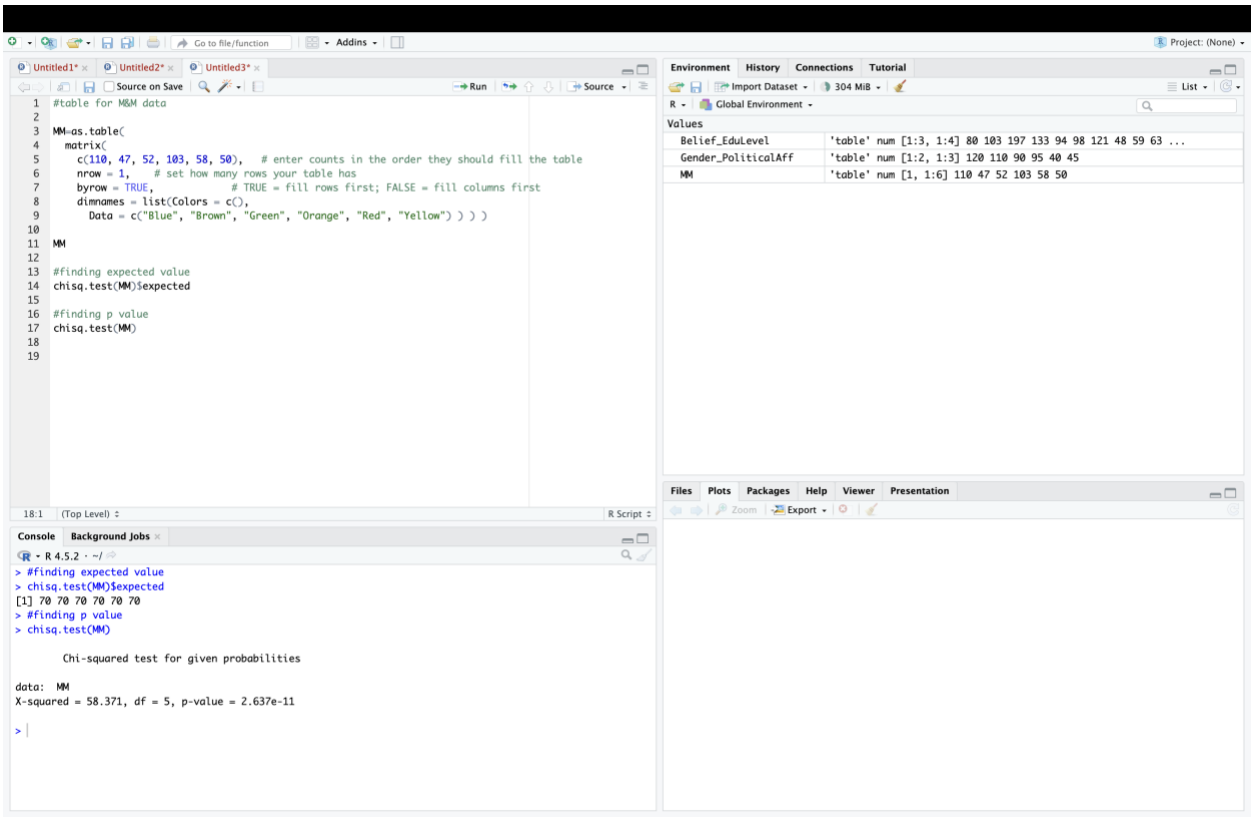
```
> #finding the expected value
> chisq.test(Belief_EduLevel)$expected
             ColumnVariable
Beliefs       HS or Less Some College College Grad Post Grad
  Believe      148.19253    126.74361     88.91552  33.14833
  Do Not Believe  94.81336     81.09037     56.88802  21.20825
  No opinion     136.99411    117.16601     82.19646  30.64342
> #finding the p-value of the data
> chisq.test(Belief_EduLevel)

        Pearson's Chi-squared test

data:  Belief_EduLevel
X-squared = 127.45, df = 6, p-value < 2.2e-16

>
```

Problem 3:


#table for M&M data
MM=as.table(
matrix(
c(110, 47, 52, 103, 58, 50), # enter counts in the order they should fill the table
nrow = 1, # set how many rows your table has
byrow = TRUE, # TRUE = fill rows first; FALSE = fill columns first
dimnames = list(Colors = c(),
Data = c("Blue", "Brown", "Green", "Orange", "Red", "Yellow"))))
MM
#finding expected value
chisq.test(MM)$expected
#finding p value
chisq.test(MM)
> #finding expected value
> chisq.test(MM)$expected
[1] 70 70 70 70 70 70
> #finding p value
> chisq.test(MM)
Chi-squared test for given probabilities
data: MM
X-squared = 58.371, df = 5, p-value = 2.637e-11
Belief_EduLevel 'table' num [1:3, 1:4] 80 103 197 133 94 98 121 48 59 63 ...
Gender_PoliticalAff 'table' num [1:2, 1:3] 120 110 90 95 40 45
MM 'table' num [1, 1:6] 110 47 52 103 58 50